First Pandas HW

This homework will take some time.

- In a markdown cell write down the AI model you use if needed
- This Homework is following the notes. Use the notes!!!!
- You will see each question and my answers below. Your HW should look very similar but with code and some more epxlanation.
- If you get stuck try the notes. Try and find the cell in the notes where we did something similar.
- If you are stuck then, use google or AI for help. If you use help on a question write down what you asked and what you leared
- The last question will have an AI component.
- I am not sure how long this HW will take.

AI Model:

Q1

- Read in the Fall2026ClassList.xlsx with sheet 'class list' and set the index_col to uni.
- Your uni should be in the class list.
- Use your pandas nomenclature to show the values associated with your uni.
- This will give you the data you need to analyze on this and future homeworks.
- You can look in the excel file to see how this is set up. But you need to do the pandas call to see it also.

Out[19]:
```
Name            Mailloux, Brian
WellData                  Drink
NYCTree             root_other
Core#                18NYHC044
Poster Group                 4
Name: bjm2103, dtype: object
```

State what you will analyze. I am looking at the parameter drink. NO ONE in the class will analyze drink. You will use one of the other columns

Q2

1. Read in the well data from EXCEL! well_data.xlsx
2. Print columns
3. show the info() for all the columns
4. Describe the data
5. make a boxplot of Arsenic split up by your parameter. Don't forget to label your y-axis and add a figure caption

You always do these steps to "see" and "understand" your data

B) show all the column names

Out[6]:
```
Index(['Well_ID', 'Lat', 'Lon', 'Depth', 'Drink', 'Aquifer', 'Shared', 'Pump',
       'Platform', 'Si', 'P', 'S', 'Ca', 'Fe', 'Ba', 'Na', 'Mg', 'K', 'Mn',
       'As', 'Sr', 'F', 'Cl', 'SO4', 'Br'],
      dtype='str')
```

C) show info()

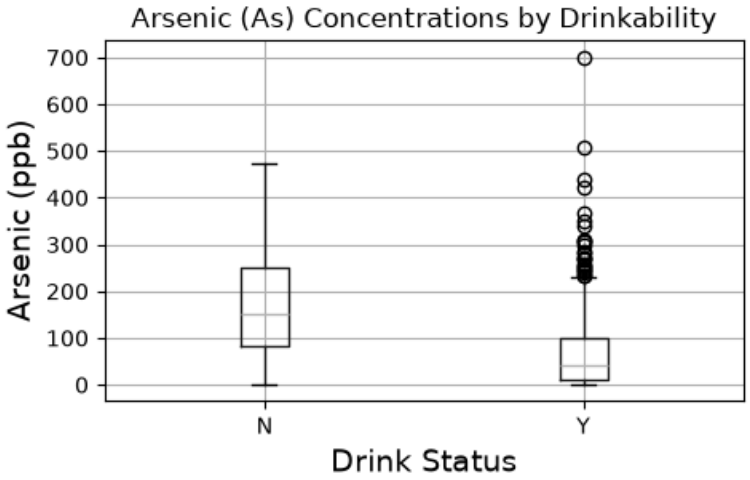
```
<class 'pandas.DataFrame'>
RangeIndex: 759 entries, 0 to 758
Data columns (total 25 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Well_ID   759 non-null    int64
 1   Lat       759 non-null    float64
 2   Lon       759 non-null    float64
 3   Depth     759 non-null    int64
 4   Drink     758 non-null    str
 5   Aquifer   759 non-null    str
 6   Shared    759 non-null    str
 7   Pump      759 non-null    str
 8   Platform  759 non-null    str
 9   Si        407 non-null    float64
 10  P         407 non-null    float64
 11  S         407 non-null    float64
 12  Ca        407 non-null    float64
 13  Fe        407 non-null    float64
 14  Ba        407 non-null    float64
 15  Na        407 non-null    float64
 16  Mg        407 non-null    float64
 17  K         343 non-null    float64
 18  Mn        407 non-null    float64
 19  As        407 non-null    float64
 20  Sr        407 non-null    float64
 21  F         413 non-null    float64
 22  Cl        411 non-null    float64
 23  SO4       397 non-null    float64
 24  Br        405 non-null    float64
dtypes: float64(18), int64(2), str(5)
memory usage: 148.4 KB
```

D) Describe the data

Out[8]:

	Well_ID	Lat	Lon	Depth	Si	P	S	
count	759.000000	759.000000	759.000000	759.000000	407.000000	407.000000	407.000000	407.0
mean	6417.088274	23.789249	90.641199	65.554677	40101.151444	0.809323	3407.292389	41129.
std	6695.778189	0.578493	0.578800	42.186161	10117.680290	0.902860	5364.247733	20161.
min	2.000000	22.780000	89.610000	0.000000	12605.576700	0.008210	-41.390000	3577.
25%	4116.000000	23.285000	90.155000	45.000000	33200.310900	0.151957	149.635000	26996.
50%	5928.000000	23.790000	90.650000	50.000000	40021.490000	0.507850	1220.877945	40166.8
75%	8134.500000	24.300000	91.130000	70.000000	45369.825000	1.189271	4341.695000	52976.4
max	141499.000000	24.770000	91.650000	523.000000	70304.057950	5.477616	45035.460000	116040.6

e) boxplot



Q3

Use value counts to see how the values of your paramter. Describe in one sentence.

```
Drink
Y    614
N    144
Name: count, dtype: int64
```

Q4

Use groupby and describe to summarize the arsenic compared to your parameter. Compare in 1-2 sentences

Out[7]:

Drink	count	mean	std	min	25%	50%	75%	max
N	71.0	171.105792	107.308224	1.368709	81.614239	150.250	250.245000	473.34
Y	336.0	72.484421	91.571489	0.000000	9.962403	39.975	99.294435	700.89

Q5

Use a t-tetst to see if the mean arsenic concentrations between your groups are signifcant. Say in 1 sentence if signicant or not and at what level.

Out[8]: `TtestResult(statistic=-7.209206229150192, pvalue=1.48295794648615e-10, df=92.74138989735245)`

The means between ..... are ..... at the level of p<0.001.

Q6

- Plot Arsenic versus depth.

- Add a 1 sentence figure caption.
- Ask AI or google for help. You will need to search how to invert your y-axis (depth always goes down) and how to get the ticks and labels to the top. Inverting is an easy search. For the next part you need to search how to do labels and how to do ticks.
- Also for depth plots making them narrower and taller to give a sense of depth.

Out[20]: `Text(0, 0.5, 'Depth (ft)')`

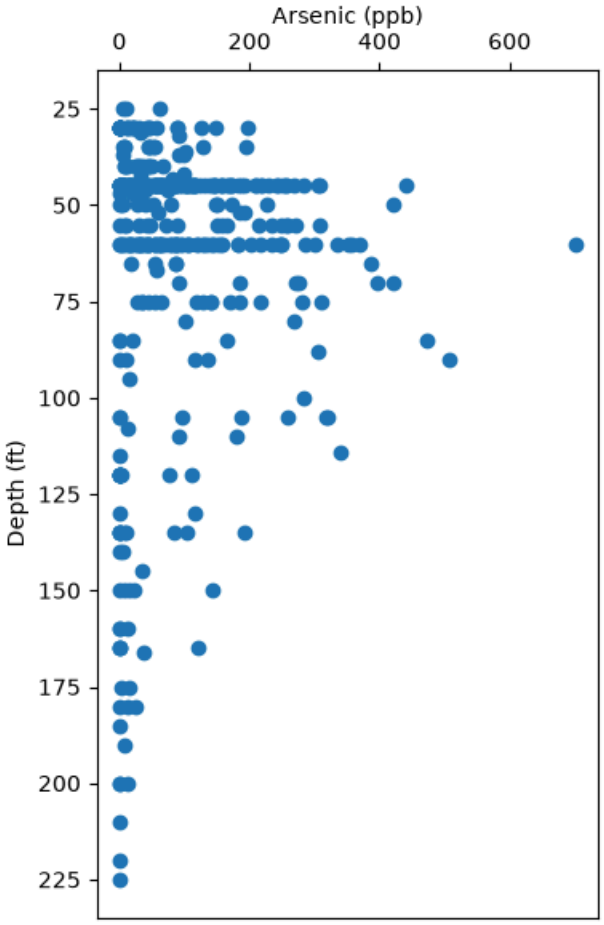


Figure Q6. Plot of .....

Q7

- Plot arsenic Versus Depth colored by your parameter.
- Choose your favorite colors. A
- dd a 1 sentence figure caption.
- Some hints.
  - Copy your plot from above.
  - But now you will call scatter twice. Once for each value of your parameter.
  - You know from the packet how to look at a subset of the data.
  - Make sure you do that for both depth and arsenic. If you do not you get an error saying "x and y must be the same size".
  - add the labels.
  - add the color.
  - then add the legend.

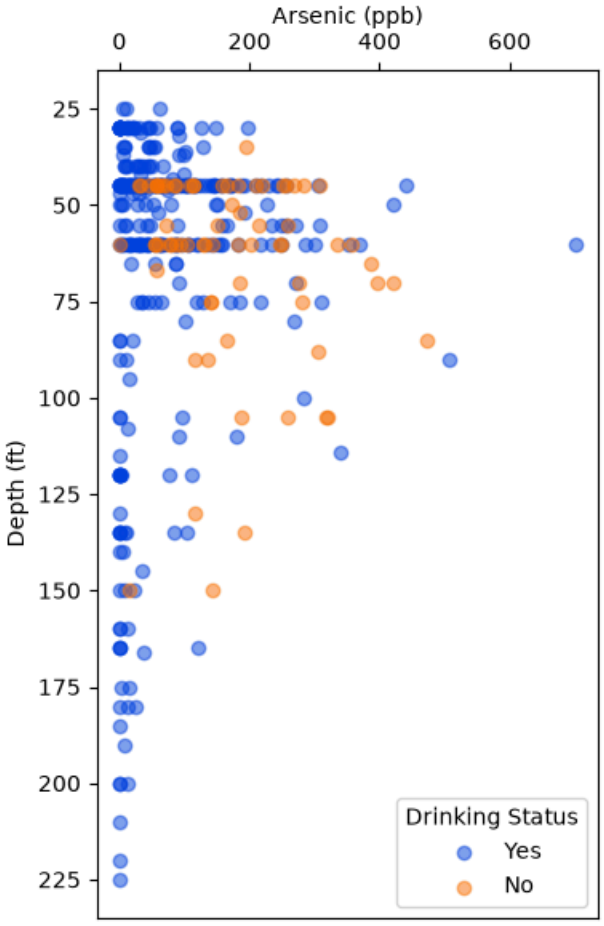
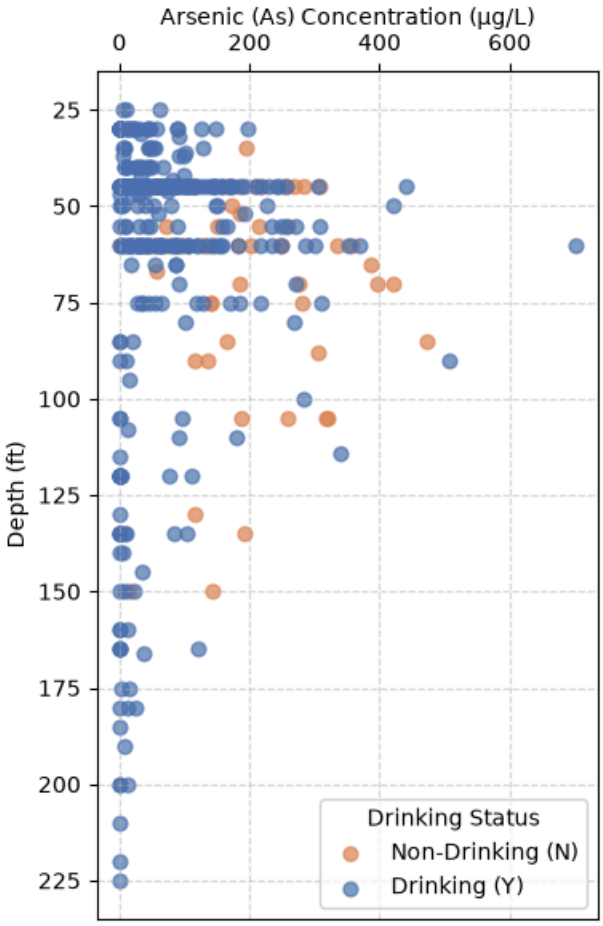


Figure Q7. Plot of .....

Q7 AI

Even if you got Q7 correct you need to use AI to make the same figure 7. In markdown write what you will paste into AI. Then keep tweaking until you get the figure out want. AI will probably do a groupby for loop for the plotting. I did two calls to scatter. Both are good ways. We will get to groupby later in the semester. If you don't get the plot you want alter your bullet points in markdown, paste them in again and see what you get. You need to get to a good graph! Make AI work for you.

- Can you read in the file well_data.xlsx and set the index to Well_ID
- Plot ....
- use ......



AI Comments? Anything learned?